Week 9: 26th- 30th October 2026

Subject / الموضوع: AP Chemistry **Unit Title / عنوان الوحدة: Units 6–7 Thermochemistry & Equilibrium** **Grade / الصف: 12**

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Session 1 الحصة الأولى **6.9 Hess's Law** قانون هس	**DOK 3 – Analyze:** I can represent a reaction as a sequence of steps and use Hess's law to combine the enthalpies of those steps to calculate ΔH for the overall reaction, justifying each change I make to the equations.	3.3 Guide processing with explicit steps, chunking, and interactive models. 2.2 Clarify syntax/structure and highlight relationships in text, formulas, diagrams. 5.3 Build fluency with graduated scaffolds and models. **Formative:** Hess's law practice problems and exit ticket	Kindly check Toddle for the lesson PPT and practice sheet. AP Classroom: AP Daily video and Topic Questions for 6.9.
Session 2 الحصة الثانية **Unit 6 Progress Check (FRQ)** التقييم التكويني للوحدة السادسة	**DOK 3 – Apply and justify:** I can apply Unit 6 thermochemistry concepts to solve multi-step, AP-style free-response questions and justify my answers with calculations and scientific reasoning.	7.3 Minimize threats and distractions through predictable structures and varied stimulation. 4.2 Ensure access to tools and assistive technologies. 8.4 Provide mastery-oriented feedback that emphasizes growth and improvement. **Graded formative:** College Board Unit 6 Progress Check (written FRQ) on AP Classroom	AP Classroom: Unit 6 Progress Check. AP Chemistry equations and constants sheet.
Session 3 الحصة الثالثة **7.1 Introduction to Equilibrium** مقدمة في الاتزان الكيميائي	**DOK 3 – Explain with evidence:** I can use experimental observations and concentration–time graphs to explain how a reversible reaction reaches dynamic equilibrium.	2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 3.2 Highlight big ideas, patterns, and critical features using organizers and examples. 8.3 Foster collaboration and community with structured group roles and supports.	Kindly check Toddle for the lesson PPT and practice sheet. AP Classroom: AP Daily video and Topic Questions for 7.1.
Session 4 الحصة الرابعة **7.2 Direction of Reversible Reactions** اتجاه التفاعلات العكسية	**DOK 3 – Analyze:** I can analyze the relative rates of the forward and reverse reactions to predict and justify the direction in which a reversible reaction proceeds.	3.1 Activate prior knowledge using organizers, links, and pre-teaching. 6.3 Facilitate organization with graphic organizers, prompts, and note-taking guides. 9.3 Develop self-assessment and self-reflection with tools, charts, and feedback displays.	Kindly check Toddle for the lesson PPT and practice sheet. AP Classroom: AP Daily video and Topic Questions for 7.2.

Homework الواجبات	• Before Session 2: review Unit 6 (6.1–6.9) and complete the 6.9 Topic Questions on AP Classroom. • After Session 2: correct the Unit 6 Progress Check using the feedback. • After Sessions 3 and 4: complete the 7.1 and 7.2 Topic Questions on AP Classroom.	3.4 Support transfer using checklists, organizers, and recurring practice. 8.2 Vary challenge level with differentiated tools, resources, and difficulty. 6.4 Support progress monitoring with rubrics, checklists, and visual progress tools.	AP Classroom and Toddle
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	Session 2 is a graded formative assessment: the College Board Unit 6 Progress Check (written free-response questions) on AP Classroom. Students need a calculator and the AP equations and constants sheet. Session 1 completes Unit 6 Thermochemistry, and Session 3 begins Unit 7 Equilibrium.		

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Technology Integration دمج التكنولوجيا	AP Classroom (AP Daily videos, Topic Questions and the Unit 6 Progress Check), Toddle for lesson materials, and graphing tools for concentration–time and rate–time graphs.
Cross-curricular links الروابط بين المواد الدراسية	Mathematics (rearranging and combining equations, interpreting graphs) and Physics (conservation of energy and state functions).
External Assessment Integration دمج التقييمات الخارجية	AP Chemistry Exam (College Board): Course and Exam Description Units 6 and 7; Unit 6 Progress Check on AP Classroom.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	Self-management (preparing for and staying calm during the progress check), thinking (analysing multi-step problems) and communication (justifying answers in writing, as on the AP free-response section).
Standards المعايير	• **6.9 Hess's Law – ENE-3.E:** Represent a chemical or physical process as a sequence of steps. • **6.9 Hess's Law – ENE-3.F:** Explain the relationship between the enthalpy of a chemical or physical process and the sum of the enthalpies of the individual steps. • **7.1 Introduction to Equilibrium – TRA-6.A:** Explain the relationship between the occurrence of a reversible chemical or physical process, and the establishment of equilibrium, to experimental observations. • **7.2 Direction of Reversible Reactions – TRA-6.B:** Explain the relationship between the direction in which a reversible reaction proceeds and the relative rates of the forward and reverse reactions.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	**Inquiry:** How can we find the energy change of a reaction we cannot measure directly, and what decides which way a reversible reaction goes? **TOK:** If we cannot see particles reacting, how do we know that equilibrium is dynamic rather than static?
Success Criteria معايير النجاح **(Measurable)**	The student has met the learning targets by: لقد حقق الطالب/الطالبة أهداف التعلم من خلال: **Session 1 – 6.9 Hess's Law** • **MUST (DOK 1):** I can state Hess's law and identify that reversing an equation changes the sign of ΔH and multiplying an equation multiplies ΔH by the same factor. • **SHOULD (DOK 2):** I can combine two or three given equations to calculate ΔH for a target reaction, showing every step. • **COULD (DOK 3):** I can justify why ΔH does not depend on the pathway (enthalpy is a state function) and solve multi-step problems with four or more equations. **Session 2 – Unit 6 Progress Check (FRQ)** • **MUST (DOK 1):** I can calculate heat using q = mcΔT and ΔH with the correct sign and units. • **SHOULD (DOK 2):** I can solve multi-step problems using Hess's law, bond enthalpies or enthalpies of formation, showing all my working.

	• **COULD (DOK 3):** I can justify my answers with particle-level reasoning and, after feedback, correct my errors and explain each misconception. **Session 3 – 7.1 Introduction to Equilibrium** • **MUST (DOK 1):** I can define a reversible reaction and dynamic equilibrium, and identify where equilibrium is reached on a concentration–time graph. • **SHOULD (DOK 2):** I can explain that at equilibrium the forward and reverse rates are equal and concentrations stay constant, but are not necessarily equal. • **COULD (DOK 3):** I can use observations, such as a colour that stops changing, to argue that a system has reached equilibrium and explain why it is dynamic, not static. **Session 4 – 7.2 Direction of Reversible Reactions** • **MUST (DOK 1):** I can identify the forward and reverse reaction rates on a rate–time graph. • **SHOULD (DOK 2):** I can predict the net direction of a reaction when the forward rate is greater than, or less than, the reverse rate. • **COULD (DOK 3):** I can justify, using how rate depends on concentration, how both rates change over time until they become equal, and sketch the rate–time graph.
National Identity Domain **مجالات الربط بالهوية الوطنية**	Conservation الحفاظ علي البيئة Global Understanding التفاهم العالمي Arabic Language اللغة العربية **Area:** Energy and sustainable industry in the UAE, linked to the UAE Net Zero by 2050 strategic initiative. **المجال** **Implementation:** **التطبيق** • Session 1: students use Hess's law to compare the enthalpy of combustion of natural gas (methane) with that of hydrogen, and discuss why cleaner fuels matter for the UAE's Net Zero 2050 goal. • Sessions 3–4: students explore the reversible reaction that makes ammonia, as produced in Ruwais, and discuss how controlling the direction of a reaction supports UAE industry. • Arabic key terms: المحتوى الحراري (enthalpy), قانون هس (Hess's law), تفاعل عكسي (reversible reaction), الاتزان الكيميائي (chemical equilibrium), سرعة التفاعل (reaction rate).